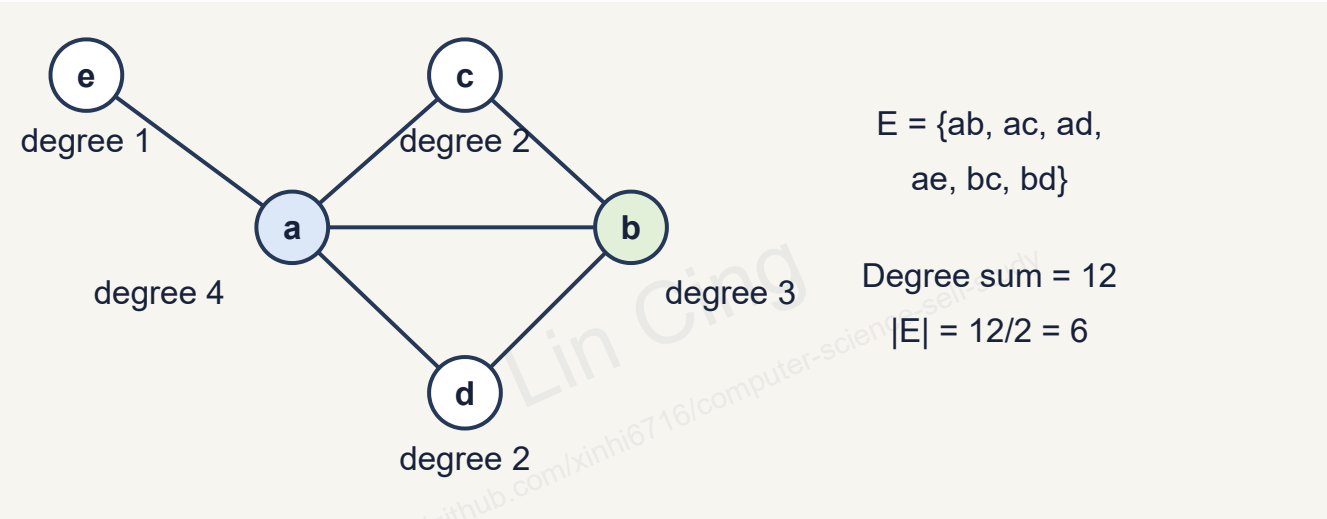
Graph-theory solutions

1

The degree sum is twice the edge count and thus even. Even-degree vertices contribute an even sum, so odd-degree vertices also have an even sum. A sum of s odd integers has parity s; therefore s is even.

2

The sum is twelve, so six edges are required. Havel–Hakimi first connects degree-four vertex to all four others, leaving sorted residual degrees (2,1,1,0). Connect degree two to the two positive degrees, leaving zeros. Explicit vertices a, b, c, d, e and edges ab, ac, ad, ae, bc, bd give degrees 4,3,2,2,1.



3

The sum is 2・3+6・4=30=2e, giving e=15. Connectivity and exactly two odd vertices guarantee an Euler trail with those degree-three endpoints. There is no Euler circuit.

4

$K_{3,3}$ is simple bipartite and has no cycle shorter than four. A planar embedding would therefore satisfy $4f \leq 2e$, counting each edge side once. It is connected, so Euler's formula with v=6,e=9 gives f=5. But 4f=20>18=2e, contradiction. This excludes every embedding.

Graph-theory solutions (continued)

5

Name cyclic vertices v_1,v_2,v_3,v_4 and split by opposite v_1,v_3 colors. Equal colors give k choices for that color and k−1 choices independently for each of v_2,v_4: $k(k-1)^2$. Distinct colors give $k(k-1)$ ordered choices and $(k-2)^2$ for the others. Hence

$$P_{C_4}(k) = k(k-1)^2 + k(k-1)(k-2)^2 = k(k-1)(k^2-3k+3).$$

At k=2 it gives two alternating colorings, as expected for this connected bipartite graph.

6

Deleting a bridge separates two sides. A cycle using it would need another edge crossing back between the sides, contradicting that it is the unique cut edge. Thus no Hamilton cycle uses a bridge.

After connecting two K_5 graphs, n=10. Bridge endpoints have degree five, but the other eight vertices have degree four<5 and fail Dirac's universal threshold. In each clique choose a Hamilton path ending at its bridge endpoint; concatenate across the bridge to visit all ten vertices once. A Hamilton cycle would need to cross between cliques twice, impossible with the only crossing edge being that bridge. This is not a counterexample to Dirac; its degree premise fails.

7

Both have V={A, B, C, D}, E={AB, AC, AD, BC, BD, CD}. The crossing of AC, BD is not in V and does not split either edge, so there remain four vertices and six edges. The crossed drawing alone proves nothing about nonplanarity, while the right crossing-free drawing proves K_4 planar.

Initially (v, e, f)=(4,6,4). AD is not a bridge because A–B–D remains; deleting it merges F_1,F_3, giving (4,5,3). BD is then not a bridge because B–C–D remains; deletion merges inner faces, giving (4,4,2). AC still has alternate A–B–C, so deletion gives (4,3,1) by merging inner and outer faces. Each step preserves v−e+f=2. Remaining AB, BC, CD form a tree. CD is now a bridge; deleting it would isolate D and cannot use the two-face merge argument.

For $K_3 \sqcup K_2$, v=5,e=4,c=2. The triangle encloses one face and all components share the unbounded face, so f=2. Then 5−4+2=3=1+c. Separate outer-face counts would duplicate one geometric region.

Graph-theory solutions (continued)

8

In A, B, C, D order both outdegrees and indegrees are 2,1,2,1. The nonisolated underlying graph is connected and every vertex balanced. A circuit is A→B→C→A→C→D→A, using AB, BC, CA, AC, CD, DA once each. There is no open Euler trail with different endpoints, since no vertices have imbalances +1,−1.

Adding X→Y→Z→X leaves old degrees unchanged and gives each new vertex indegree/outdegree one. Balance remains but nonisolated underlying connectivity fails: two components cannot be covered by one trail. All nine arcs cannot share a circuit. Both degree and connectivity conditions are essential.

9

There are 6 · 7=42 vertices. Each has four distinct neighbors (i±1,j),(i, j±1), giving degree four and e=42 · 4/2=84.

Every step changes exactly one coordinate. In a three-step closed walk changing both coordinates, one coordinate changes exactly once; ±1 cannot vanish modulo six or seven. If all three steps change one coordinate, the total change belongs to {−3,−1,1,3}, again nonzero modulo either. Thus no triangle exists. The graph is simple and connected, since coordinates can be adjusted consecutively to reach any vertex.

A triangle-free planar graph would have e≤2v−4=80, contradicting 84. The graph actually has a seven-cycle in the second-coordinate direction and is not bipartite. The proof uses triangle-freeness, not a false claim that every cycle is even.

10

Five vertices have ten possible edges, so exactly four gives $\binom{10}{4} = 210$. Inclusion–exclusion for no isolates gives

$$2^{10} - 5 \cdot 2^6 + 10 \cdot 2^3 - 10 \cdot 2 + 5 - 1 = 1024 - 320 + 80 - 20 + 5 - 1 = 768.$$

For connectivity use c_1=1,c_2=1,c_3=4,c_4=38 and classify the component of vertex one. Disconnected contributions at k=1,2,3,4 are

$$\binom{4}{0} \cdot 1 \cdot 2^6 = 64, \quad \binom{4}{1} \cdot 1 \cdot 2^3 = 32, \quad \binom{4}{2} \cdot 4 \cdot 2 = 48, \quad \binom{4}{3} \cdot 38 = 152.$$

Thus c_5=1024−(64+32+48+152)=728. The difference forty consists of no-isolate disconnected graphs with component sizes 2+3. Choose the unique two-vertex component in ten ways; it has one edge, and the remaining three vertices have four connected graphs, giving 10 · 4=40. No division by two is needed because the different component sizes distinguish them uniquely.

Graph-theory solutions (continued)

11

Removing |S| vertices from a Hamilton path leaves at most |S|+1 path segments. Other edges in G−S can only merge them, proving c(G−S)≤|S|+1, including empty S, which requires original connectivity.

For $K_{2,4}$ delete the two-vertex part. Four isolates remain, violating 4≤3, so no Hamilton path exists. Equivalently an alternating path cannot use part sizes differing by more than one.

$K_{3,4}$ has path $b_1–a_1–b_2–a_2–b_3–a_3–b_4$. Every edge crosses parts and each vertex occurs once. A cycle would alternate equally many vertices from both parts, impossible for 3≠4. Alternatively deleting the three-vertex part leaves four components, violating the Hamilton-cycle bound.

12

Sequential coloring gives P_C_3(k)=k(k−1)(k−2). Deleting an edge of C_4 gives a four-vertex tree; contracting it gives C_3:

$$P_{C_4} = k(k-1)^3 - k(k-1)(k-2) = k^4 - 4k^3 + 6k^2 - 3k.$$

For C_5 deletion gives a five-vertex tree and contraction C_4:

$$P_{C_5} = k(k-1)^4 - P_{C_4} = k^5 - 5k^4 + 10k^3 - 10k^2 + 4k.$$

At two, 32−80+80−40+8=0: an odd cycle is not two-colorable. At three, 243−405+270−90+12=30 proper colorings with three named colors exist. Every one actually uses all three, since using fewer would give a two-coloring. Thus $\chi(C_5)=3$; thirty counts colorings, not the minimum number of colors.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study